

END SEMESTER ASSESSMENTS, SPRING 2024

Program: B.TECH COMPUTER SCIENCE & ENGINEERING/ B.TECH COMPUTER SCIENCE & BUSINESS SYSTEMS/ B.TECH COMPUTER SCIENCE & ENGINEERING (LATERAL)/ B.TECH COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)/ B.TECH COMPUTER SCIENCE AND ENGINEERING (IOT, CYBERSECURITY AND BLOCKCHAIN TECHNOLOGY)	
Subject Name: ALGORITHM II	
Subject Code: 1000013102/ 1000023102/ 1000033102/ 1000103102/ 1000113102	Semester: IV
Duration: 2 Hours	Maximum Marks: 50

Note: Answer the following questions as directed.

Group A

(Answer All questions)

(Each question carries 1 mark)

1. Choose the correct answer from the options given: (10 X 1)

(i) Which of the following is a correct interpretation of $O(n)$ as amortized time complexity?

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|---|--|
| (a) Each operation takes constant time on an average. | (b) Each operation takes constant time in the worst-case scenario. |
| (c) The total time for a sequence of operations is linear in the size of the input. | (d) Each operation takes logarithmic time. |

(ii) Topological sorting is primarily used for

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|--|---|
| (a) Finding the shortest path between any two vertices in a graph. | (b) Detecting cycles in a directed graph. |
| (c) Finding an order of task execution from the task precedence graph. | (d) Constructing minimum spanning trees. |

(iii) The minimum time complexity for evaluating an n -degree polynomial is

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|-----------------|------------|
| (a) $O(n^2)$ | (b) $O(1)$ |
| (c) $O(\log n)$ | (d) $O(n)$ |

- (iv) In the Ford-Fulkerson algorithm, the residual capacity of a given edge is determined
- (a) By subtracting the flow from its capacity.
 - (b) By adding the flow to the capacity.
 - (c) By dividing the flow by the capacity.
 - (d) By multiplying the flow with the capacity.
- (v) In executing the FFT algorithm, the number of sample points used in the two recursive calls at any level of recursion will be equal, when the total number of sample points is
- (a) Prime
 - (b) Even
 - (c) Odd
 - (d) Some Power of 2
- (vi) Which of the following algorithms is commonly used to compute the convex hull of a set of points on a plane?
- (a) Breadth-first search
 - (b) Dijkstra's algorithm
 - (c) Graham scan
 - (d) Depth-first search
- (vii) Which statement is true regarding the complexity of the Traveling Salesman Problem?
- (a) It is a polynomial-time problem
 - (b) It is an NP-hard problem
 - (c) It can always be solved optimally using greedy algorithms
 - (d) It has a fixed time complexity for all instances
- (viii) What is the worst-case time complexity of the KMP algorithm for searching a pattern of length m in a text of length n ?
- (a) $O(n)$
 - (b) $O(m)$
 - (c) $O(n + m)$
 - (d) $O(n * m)$
- (ix) How does the KMP algorithm avoid unnecessary comparisons during string matching?
- (a) By using a sliding window approach
 - (b) By skipping positions in the text based on the longest prefix-suffix information
 - (c) By dividing the text into smaller blocks
 - (d) By utilizing a hash table for pattern matching

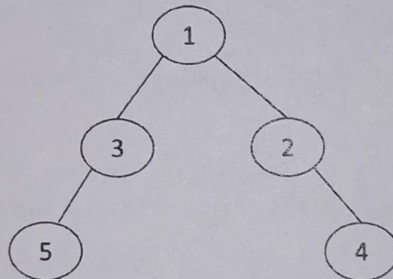
- (x) Batcher's odd-even merge sort can be efficiently implemented in parallel computing environments due to its
- | | |
|--------------------------|-----------------------------|
| (a) Sequential nature | (b) High memory requirement |
| (c) Inherent parallelism | (d) Limited scalability |

Group B

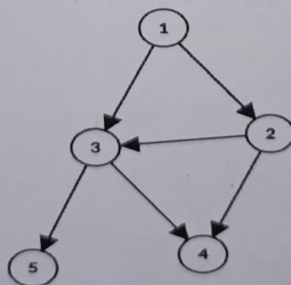
(Answer any Four questions out of Six)
(Each question carries 5 marks)

(4 X 5)

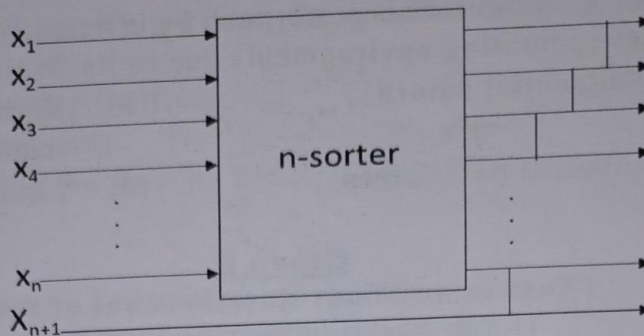
2. Trace the execution of link inversion preorder traversal of the following binary tree, considering *pres* as the present node and *prev* as the previous node.



3. Compute the linear order of the following graph using topological sort. Show all steps of the algorithm used by you.



4. Consider the following schematic diagram of a sorting network for sorting $(n + 1)$ elements. Based on this scheme, draw the sorting network for sorting $n = 5$ elements and find its parallel running time for sorting.



5. Show how two polynomials, each of degree n can be multiplied in $O(n \log n)$ time.
6. Consider two problems P_1 and P_2 where P_1 is known to be NP-complete. Describe the steps to show that P_2 is also NP-complete.
7. Describe the algorithm for Travelling Salesman Problem.

Group C

(Answer any Two questions out of Four)
(Each question carries 10 marks)

(2 X 10)

8. Write down the algorithm to perform in order traversal using link inversion in a binary tree.
9. Draw the network for sorting 8 elements using Batcher's odd-even merge sort algorithm. Show all the necessary steps.
10. Compute the number of comparators required to sort $n = 2^k$ elements using Batcher's odd-even merge sort algorithm. Show all steps in detail.
11. Compute the maximum flow of the following network using Ford-Fulkerson algorithm,

